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Appendix

A Appendix of Proofs

A.1 Proof of Proposition 1

Case $b < 1$ and $d = 1$: Suppose $a^*(s^E; p) = 0$ and $a^*(s^A, s^E; p) = 1$. Equivalently, $\log c_{rel} > \log \frac{\pi(\omega=1|s^E)}{\pi(\omega=0|s^E)}$ and $\log c_{rel} \leq b \log \frac{\pi(s^A|\omega=1, s^E)}{\pi(s^A|\omega=0, s^E)} + \log \frac{\pi(\omega=1|s^E)}{\pi(\omega=0|s^E)}$. Since $b \in (0, 1)$, it must be that $\log \frac{\pi(s^A|\omega=1, s^E)}{\pi(s^A|\omega=0, s^E)} > 0$ and $\log c_{rel} < \log \frac{\pi(s^A|\omega=1, s^E)}{\pi(s^A|\omega=0, s^E)} + \log \frac{\pi(\omega=1|s^E)}{\pi(\omega=0|s^E)}$ so that $a^*(s^A, s^E; \pi) = 1$. Hence, if $0 = a^*(s^E; p) \neq a^*(s^A, s^E; p)$ then $a^*(s^A, s^E; p) = a^*(s^A, s^E; \pi)$ and $V(s^E; p) \leq V(s^A, s^E; \pi) = V(s^A, s^E; p)$, with strict inequality if the measure on (s^A, s^E) under $\pi(\cdot)$ such that $0 = a^*(s^E; p) \neq a^*(s^A, s^E; \pi)$ is strictly positive. The proof of the case when $a^*(s^E; p) = 1$ and $a^*(s^A, s^E; p) = 0$ is analogous. If $a^*(s^E; p) = a^*(s^A, s^E; p)$ then $V(s^E; p) = V(s^A, s^E; p)$.

Case $b > 1$ and $d = 1$: If $\log c_{rel} - \log \frac{\pi(\omega=1|s^E)}{\pi(\omega=0|s^E)} > 0$, then for $\log \frac{\pi(s^A|\omega=1, s^E)}{\pi(s^A|\omega=0, s^E)}$ $\in \left(\frac{1}{b} \log c_{rel} - \frac{1}{b} \log \frac{\pi(\omega=1|s^E)}{\pi(\omega=0|s^E)}, \log c_{rel} - \log \frac{\pi(\omega=1|s^E)}{\pi(\omega=0|s^E)} \right)$ we have both $\log c_{rel} > \log \frac{\pi(s^A|\omega=1, s^E)}{\pi(s^A|\omega=0, s^E)} + \log \frac{\pi(\omega=1|s^E)}{\pi(\omega=0|s^E)}$ and $\log c_{rel} < b \log \frac{\pi(s^A|\omega=1, s^E)}{\pi(s^A|\omega=0, s^E)} + \log \frac{\pi(\omega=1|s^E)}{\pi(\omega=0|s^E)}$. Thus, $0 = a^*(s^E; p) \neq a^*(s^E, s^A; p) \neq a^*(s^E, s^A; \pi)$. An analogous argument for the case when $\log c_{rel} \leq \log \frac{\pi(\omega=1|s^E)}{\pi(\omega=0|s^E)}$ completes this case.

Case $d \neq 1$: We analyze this in two subcases.

- $(1 - d) \log c_{rel} > 0$: We show that there exist values of $\left(\log \frac{\pi(\omega=1|s^E)}{\pi(\omega=0|s^E)}, \log \frac{\pi(s^A|\omega=1, s^E)}{\pi(s^A|\omega=0, s^E)} \right)$ such that $a^*(s^A, s^E; p) = 0$, $a^*(s^E; p) = 1$, and $a^*(s^A, s^E; \pi) = 1$. Equivalently, we need to find values such that $\log c_{rel} > b \log \frac{\pi(s^A|\omega=1, s^E)}{\pi(s^A|\omega=0, s^E)} + d \log \frac{\pi(\omega=1|s^E)}{\pi(\omega=0|s^E)}$, $\log c_{rel} < \log \frac{\pi(\omega=1|s^E)}{\pi(\omega=0|s^E)}$ and $\log c_{rel} < \log \frac{\pi(s^A|\omega=1, s^E)}{\pi(s^A|\omega=0, s^E)} + \log \frac{\pi(\omega=1|s^E)}{\pi(\omega=0|s^E)}$ if $d \neq 1$. Re-write this system as $y = \log \frac{\pi(\omega=1|s^E)}{\pi(\omega=0|s^E)} - \log c_{rel}$ and $x = \log \frac{\pi(s^A|\omega=1, s^E)}{\pi(s^A|\omega=0, s^E)}$, we need to find a solution to the system $y > 0$, $x + y > 0$ and $bx + dy < (1 - d) \log c_{rel}$. Since $(1 - d) \log c_{rel} > 0$, there exist small enough values of $x, y > 0$ such that the solution exists.
- $(1 - d) \log c_{rel} < 0$: An argument analogous of case 1 shows that there exist values of $\left(\log \frac{\pi(\omega=1|s^E)}{\pi(\omega=0|s^E)}, \log \frac{\pi(s^A|\omega=1, s^E)}{\pi(s^A|\omega=0, s^E)} \right)$ such that $a^*(s^A, s^E; p) = 1$, $a^*(s^E; p) = 0$, and $a^*(s^A, s^E; \pi) = 0$.

A.2 Proof of Proposition 2

Consider $\log c_{rel} > \log \frac{\pi(\omega=1|s^E)}{\pi(\omega=0|s^E)}$ and $\log c_{rel} < b \log \frac{\pi(s^A|\omega=1)}{\pi(s^A|\omega=0)} + d \log \frac{\pi(\omega=1|s^E)}{\pi(\omega=0|s^E)}$ so that $0 = a^*(s^E; p) \neq a^*(s^A, s^E; p) = 1$. For small enough $\log \frac{\pi(s^A|\omega=1, s^E)}{\pi(s^A|\omega=0, s^E)}$, $\log c_{rel} < \log \frac{\pi(s^A|\omega=1)}{\pi(s^A|\omega=0)} + \log \frac{\pi(\omega=1|s^E)}{\pi(\omega=0|s^E)}$ so that $a^*(s^A, s^E; \pi) = 0$. The case in which $\log c_{rel} < \log \frac{\pi(\omega=1|s^E)}{\pi(\omega=0|s^E)}$ is analagous.